

Lalit Negi, M.Tech

Member Research Staff (MRS) at Bharat Electronics Limited

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BIOGRAPHY

My primary research interests lie at the intersection of LGBTQ+ rights and social justice, Geographic Information Systems (GIS), spatial analysis, simulation, Machine Learning (ML), GeoAI, and Augmented/Virtual Reality (AR/VR). I am particularly interested in developing spatial and computational approaches to analyse the current state of LGBTQ+ rights and inclusion and to understand how legal, social, cultural, economic, and environmental factors interact across places and over time. This includes mapping disparities in rights and protections, identifying spatial patterns of discrimination and access to LGBTQ+-supportive services and spaces, and examining how different environments influence safety, belonging, participation, and quality of life. I am also interested in process-oriented approaches that examine the barriers an LGBTQ+ person may encounter at different stages of life, as well as outcome-oriented approaches that measure whether places and institutions become more accessible, safe, equitable, and inclusive. GIS, simulation, GeoAI, 3D GIS, immersive visualization, digital twins, and AR/VR can provide tools for representing these complex relationships and exploring how changes in policies, services, environments, and social conditions could influence LGBTQ+ inclusion. My previous work in Applied Cryptography and Information Security also provides a foundation for addressing privacy, security, and responsible handling of sensitive LGBTQ+ data.

EXPERIENCE

Central Research Laboratory, Bharat Electronics Limited

India, Ghaziabad

Member Research Staff

2022 – Present

GIS Tech Lab

Jan 2024- Present

- Currently working on web based **CIDSS** Command Information and Decision Support System for the Indian Army using the **Luciad RIA sdk package**.
- Leading the team working on the development of **VR Based Digital Sandtable** prototype using the **ArcGIS sdk** in Unity
- Designed a virtual Tactical Theatre Command system using **ArcGIS**, visualizing real-world deployment of defence resources, operational assets, and live tracks to support situational awareness and command-level planning.
- Successfully participated in the demonstration of the **Tactical Theatre Command System** to the **Honourable Defence Minister of India**, showcasing real-time visualization, operational planning, and decision-support capabilities.

Decision Support System Lab

July 2022 - Dec 2023

- Contributed to the development of the **first indigenously developed universal train driving simulator** through simulation algorithm design and secure communication protocol development.
- Developed simulation modules such as **Simulation Engine**, **Training and Evaluation Assessment Module**, and **Malfunction Handler** in **C++** using **Qt** to model metro train logic, control, and operational behavior.
- Prepared technical documentation including **Software Requirements Specification (SRS)** and **Software Requirement Description (SRD)** to support project delivery and audits.
- Developed a **VR-based training simulator** for radar assembly and disassembly, enabling step-by-step procedural training in an immersive and safe virtual environment.
- Developed a **SR-based display system** to create a **Digital twin** of radar systems for predictive maintenance, condition monitoring, and fault analysis.

PERSONAL PROJECTS

- **Rainbow Connect**
 - Theme: A web-based communication platform designed to support LGBTQ+ communities in offices and institutions. The platform enables users to connect through text-based communication and video calling, with support for both Hindi and English. Designed as an inclusive digital space to encourage communication, connection, and peer interaction within institutional environments.
- **Indradhanush Spaces**
 - Theme: A web-based GIS application for discovering and registering LGBTQ+-inclusive spaces. Space owners can register their inclusive spaces on the platform, while users can explore nearby spaces through an interactive map. The application uses location-based visualization to display spaces around the user's current location and identify places within a selected geographic range.
- **Geographic Information System-Based Real-Time Fire Detection and Anti-Fire Simulation System**
 - Theme: A GIS-based system for real-time fire detection, spatial visualization, and fire-spread simulation. Integrates geospatial data, real-time information, and simulation models to support fire monitoring, prediction, and emergency response planning.
- **Interactive Radar Fault Management Training System with GIS-Based Visualization and Corrective Action Guidance**
 - Theme: An interactive training system that integrates GIS-based visualization with radar fault scenarios and corrective-action guidance. Designed to provide an immersive and interactive environment for fault identification, decision-making, and technical training.
- **Immersive Virtual Reality Visualization of Machine Learning-Based Weather Classification for Climate Scenario Analysis**
 - Theme: An immersive virtual reality environment for visualizing machine-learning-based weather classification results and exploring climate scenarios. Combines VR, geospatial visualization, and machine learning to provide an interactive approach to understanding complex weather and climate data.

EDUCATION

- **Netaji Subhas University of Technology** 2020 – 2022
Master of Technology (M.Tech.) in Information Technology
 - Thesis Title: Design and Analysis of Certificateless Schemes for Healthcare Environment
 - Aggregate: 81.0% (GPA 8.1/10.0)
- **Govind Ballabh Pant University of Agriculture and Technology** 2017 – 2020
Bachelor of Technology (B.Tech.) in Computer Engineering
 - Project Title: Steg! - An Android Application for Metamorphic Cryptography
 - Aggregate: 72.4% (GPA: 7.2/10.0)

PERSONAL SKILLS

Technical Skills	◦ Domain: Inclusive Technology, GIS, ARVR, Machine Learning ◦ Languages: C#, Python, C++, TypeScript ◦ Tools: Unity, React, Vite, Canvas, Latex, Github
Communication Skills	◦ Assisted in conducting laboratory sessions and tutorials, supporting students in understanding course concepts and practical implementations. ◦ Guided students with assignments, mini-projects, and problem-solving exercises. ◦ Helped in preparing teaching materials such as lecture slides, lab manuals, and tutorial sheets. ◦ Mentoring Interns in the Integrated Domain of Machine Learning and Virtual Reality.

- Organizational Skills
- Served as an **organizer** for a **five-day technical workshop sponsored** by **All India Council for Technical Education**, coordinating planning, scheduling, and execution of technical sessions.
 - Assisted in the **preparation of workshop materials**, registrations, and participant communications.
 - Acted as an **organizing member** for the **inauguration event of a DPSU Bhawan**, supporting event planning and on-site coordination.
 - **Coordinated ceremonial arrangements**, guest management, and technical arrangements for the inauguration event.
- Certifications
- **Geographical Information System** by Department of Space, India
 - **India in a Changing Climate** by Indian Institute of Technology Hyderabad, India.
 - **Artificial Intelligence for Social Good** by Microsoft.

PUBLICATIONS

- Presented **5+ papers** in International and national conferences
- Published **2 papers** in SCI indexed Journals.
- GIS Based Real time Fire Detection and Anti Fire Simulation System.(Accepted, 47th Asian Conference on Remote Sensing (ACRS) 2026)

REFERENCES

1. **Shivani Arya**
Senior Member Research Staff, CRL-BEL, Ministry of Defence, India
Email: shivaniarya@bel.co.in
Relationship: Reporting Manager
2. **Dr. Devender Kumar Yadav**
Associate Professor, NSUT, New Delhi, India
Email: dk_iitm@yahoo.co.in
Relationship: Master's Advisor